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RESEARCH

SeCOM-B: an integrated model for understanding human behaviour change in wicked socio-ecological problems

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Abstract

The COM-B model, widely adopted in behaviour change research, systematically explores and categorises the behavioural barriers and facilitators to inform intervention design. The model highlights that where the right mix of barriers and facilitators, in the broad categories of capability, motivation, and opportunity exist, a given behaviour is more likely to be enacted. However, for wicked problems, applying the COM-B model becomes challenging due to complexity, uncertainty, manageability challenges, and the interpretative opacity of the systems that influence behaviour. This paper introduces a combined framework (SeCOM-B) that integrates the Socio-ecological model (SEM) and the COM-B model, highlighting its potential application in co-designing behaviour change interventions to address wicked problems, which often involve non-scientific stakeholders and interdisciplinary team members. Drawing on three case studies of practical behaviour change projects taking place in Australia (2) and Vietnam (1) between March 2022 and July 2023, the paper further illustrates the application of the SeCOM-B model in analysing the drivers and barriers of behaviours, and exploring the implications for intervention design.

Keywords Behaviour change · COM-B model · Knowledge co-production · Socio-ecological model · Sustainability · Wicked problems

1 Introduction

One of the challenges of sustainable development is taming "wicked problems". In a review of 50 years of the concept, Chan and Xiang (2022, p.2) acknowledge that "wickedness, the ubiquity of wicked problems, is the norm in social systems of all kinds, including socio-ecological systems where human–human (social–economic–political–cultural–…) relations and human–nature (ecological) relations are intertwined with each other".

Although the term was only coined in the 1960s in English literature, the phenomenon of "wicked problems" has a much longer history (Xiang 2013, pp.1–2). The concept of wicked problems, coined by Rittel and Webber (1973, pp.160–166), is characterised by a lack of clear problem definition, an absence of definitive solutions, and the inextricable entanglement between understanding and solving the problem, with each problem itself a symptom of another underlying issue within a complex system. The complex nature of wicked problems, in which the implications of any change ripple through systems over long time periods, makes researching and designing a behaviour change intervention

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to address one part of a wicked problem extremely difficult (Hallsworth 2023).

Consider the need to understand and address the problem of climate hazards: while individual behaviours may increase or decrease harm from a specific event, to truly understand and address harm, we must look beyond the present timeframe to how disaster vulnerability (i.e. the coherent weakness of the affected community) is embedded in social, political, cultural, and economic systems, which evolve over decades to centuries (Wisner et al. 2004, 2012). As a specific example, to analyse the drought and saline intrusion disaster that damaged more than 50% of the Vietnamese Mekong Delta's total planted rice crop areas, understanding farmers' risk-taking (e.g. growing the disaster-prone crop) and risk-hedging behaviours (e.g. dropping the crop and choosing safer livelihoods) requires an understanding the interdependent systems of community networks, the agricultural system, the political regime, culture, the built environment, and the natural environment (Nguyen-Trung et al. 2023). Each of these systems has co-evolved over time and combines uniquely to influence individual risk-taking behaviour.

Behavioural science, with its focus on understanding and influencing human actions, can be used as a methodological lens (Hallsworth 2023; Kaufman et al. 2021) to navigate complex sustainability problems. For example, by asking "WHO needs to do WHAT differently" (Kneebone et al. 2024), or by identifying the *Action, Actor, Context, Target,* and *Time* components of behaviour change needed to create valued outcomes, stakeholders can discuss and agree on ways forward through complex challenges using human behaviour as a primary focus (Smith & Curtis 2020; Presseau et al. 2019). After agreeing on the correspondence between a behaviour change and a valued outcome, behavioural science methods can also be useful in identifying drivers and barriers that make a behaviour more or less likely, and in designing interventions tailored to influence that behaviour. A widely used and well-evidenced example of behavioural science is the COM-B framework (Michie et al. 2011), which posits that a specific behaviour (B) is a product of individual capability (C), opportunity (O), and motivation (M).

2 Overview of COM-B and socio-ecological models

2.1 The value and limitations of COM-B model

The COM-B model forms the core of the Behaviour Change Wheel (BCW), a framework proposed by Michie et al. (2011), which links each component (capability, opportunity, motivation) to potential behaviour change interventions. The COM-B model therefore aids in diagnosing behaviour change needs and supports the design of behaviour change interventions in diverse areas such as healthcare settings (Alexander et al. 2014; Alhusein et al. 2021; Bhandari et al. 2021), physical activity and diet (Craveiro et al. 2021; Livingstone et al. 2023; Fuster et al. 2023), risk-taking in agriculture (Irwin et al. 2022), consumer behaviours towards plant-based meat alternatives (Jiang and Farag 2023; Li and Kang 2022), public transportation (Krusche et al. 2022), sport and technologies (Creaser et al. 2022), and sustainable practices such as food waste disposal behaviours (Manika et al. 2022; Wastling et al. 2018). The COM-B model presents a "behaviour system", with each letter representing different essential conditions for behaviour to take place:

- "C" stands for capability, which includes physical and psychological capability. Physical capability pertains to bodily skills, strength, and stamina, while psychological capability refers to mental aspects such as understanding, knowledge, and the capacity to perform cognitive processes like comprehension and reasoning.
- "M" stands for motivation, which includes reflective and automatic motivation. Reflective motivation involves conscious and reflective processes, such as evaluations or plans, whereas automatic motivation involves subconscious processes and habits, including emotional reactions, impulses, and desires.
- "O" stands for opportunity, which includes physical and social opportunity. Opportunity refers to factors external to the individual that enable or restrict the behaviour, such as the physical environment (time, resources, access, cues) and social settings (norms, culture, social groups).

The COM-B model is applied in behaviour change research to analyse physical, psychological, social, and environmental influences on a target behaviour. This method works especially well when applied to the analysis of behavioural drivers and barriers at the individual (capability and motivation) level.

However, COM-B falls short when used to analyse drivers and barriers beyond the individual—especially "opportunity". In the COM-B model, both individually-experienced factors (e.g. money, time, familial norms) and complex, multi-level structural factors (e.g. government policy, societal expectations, institutional processes) are consolidated into a single category of influence, "opportunity". Michie et al. (2014, p. 51) defined "opportunity" as physical influences (such as "what the environment allows or facilitates in terms of time, resources, locations, physical barriers") and social influences (such as "interpersonal influences, social cues, and cultural norms"). Yet, the concept of opportunity is

insufficient for capturing how "wicked problems", including those in sustainable development, interact with individual behaviour, especially (a) their *complexity*, the multiplicity of causes and consequences spanning sectors and scales, which are embedded in societal structures and institutions; (b) their *uncertainty*, the absence of ready-made solutions and the way each attempted fix reshapes the problem definition; (c) their *manageability challenges*, the involvement of autonomous actors with divergent interests; and (d) their *interpretative opacity,* the ill-structured, dynamic nature that resists straightforward understanding of the systems in which behaviour change occurs (Rotmans 2005, pp. 7–8). As one example, consider the challenge of mainstreaming nature-based solutions in cities. Identifying drivers and barriers for behaviours that can address this challenge must engage with a city's multi-level governance landscape, which could include national, state, regional, and municipal influence on "what the environment allows" or "cultural norms" (Adams et al. 2023, p. 5). An individuals' likelihood to enact behaviour change is also influenced by their experience of overlapping systems such as housing, education, work, and many others (Frantzeskaki et al. 2012, p. 29) or their social environment and relationships, including with other members of their community, local businesses, political representatives, and many others (Adams et al. 2024, p. 3).

The need to integrate systems and structure into the analysis of behaviour has become more visible in the context of the "wicked problems" that often emerge on the pathways towards sustainable development. Shove (2010, p. 1274) has argued that individual-focussed analyses of behaviour, such as Ajzen's (1991) Theory of Planned Behaviour (TPB), should be abandoned in favour of "social theories of change and transition". Shove argues that the emphasis on individual human behaviour can mislead us into thinking that "wicked problems" such as climate change or disasters are ultimately due to human choices, such that if individuals change their behaviour, the problems can be fixed. Understanding an "s-frame"—"the system of rules, norms, and institutions" that shape behaviours—would allow researchers and practitioners to produce better insights when working with the "i-frame"—"individuals, their thoughts and behaviors" (Chater and Loewenstein 2022, pp.1, 14).

2.2 The values of the socio-ecological model (SEM)

Socio-ecological models (SEMs) are helpful in dealing with wicked problems since they can situate behaviour within a set of nested systems. SEMs have been employed in behavioural research in areas such as health and eating behaviours (Ma et al. 2017; Robinson 2008; Townsend and Foster 2013), sport and physical activities (Essiet et al. 2017; Vella et al. 2014), gender (Sword 1999), help-seeking and community engagement (Alhomaizi et al. 2018; Caperon et al. 2022), juvenile problem behaviours (Scholte 1992), and transportation (Acheampong and Siiba 2018).

While numerous types of socio-ecological models exist, many are adaptations of an ecological system theory of human development proposed by Urie Bronfenbrenner (1977, 1979). According to Bronfenbrenner's model (1977, p. 514), people and their behaviours are embedded within "a nested arrangement of structures, each contained within the next". These structures include *microsystems* (i.e. immediate settings including home, school, and workplace); *mesosystems* (i.e. the interaction between a person's major microsystems, such as home and school), *exosystems* (i.e. the interaction between two or more settings that do not necessarily contain the person under study but indirectly influence him/her, such as the family, the school, and the peer group), *macrosystems* (i.e. overarching institutional structures embedded in a given culture or subculture), and *chronosystems* (i.e. "change or consistency over time not only in the characteristics of the person but also of the environment in which the person lives" (Bronfenbrenner 1994, pp. 39–40).

When discussing systems, their interaction, and influence on behaviour, it is crucial to clarify the distinction between systems and structures (Giddens 1979). A system, as "social networks with mutual dependencies and ongoing interactions between actors" (Grin et al. 2010, p. 44), typically has three components: functions or purposes, elements, and interconnections between elements (Meadows 2008, p.12). The interconnections among a system's elements are governed by structures, that may take the form of *rules* and/or *resources (or capital),* which enable or constrain individuals' behaviours (Meadows 2008, p. 16; Bourdieu 1977; 1990). As elaborated by Grin et al. (2010, p. 44):

> [s]ystems and structures thus provide two types of context and embeddedness for actors, the former more horizontal, oriented towards interactions with other mutually dependent actors (e.g. exchange, conflict, coalition, negotiation, strategic games), and the latter more vertical, oriented towards formal, cognitive and normative rules. (Grin et al. 2010, p. 44)

Structures in the form of *rules* refer to mental frames that guide but do not determine actions, including three types: *cognitive rules* (e.g. belief systems, guiding principles, goals, innovation agendas, problem definitions and search heuristics), *normative rules* (role relationships, cultural norms, values and behavioural norms), and *regulative or formal rules* (regulations, standards and laws) (Grin et al. 2010, p. 20). Meanwhile, *resources* refer to potential capital that enables individuals and groups to build their capacity and obtain goals. In this context, resources or capital can

be natural, physical, economic, cultural, social, or political (Nguyen-Trung et al. 2023).

Bronfenbrenner's model and the systems approach have been combined in different ways to analyse individual behaviour. For instance, McLeroy et al.(1988, p. 355) developed a socio-ecological model for health promotion that comprises five levels: *intrapersonal factors* include individual (internal) characteristics such as knowledge, attitudes, behaviour, self-concept, and skills; *interpersonal processes and primary groups* encompass formal and informal social networks and social support systems, such as family, work groups, and friendship networks*; institutional factors* involve social institutions with their rules and regulations, both formal and informal*; community factors* refer to relationships among organisations, institutions, and informal networks within defined boundaries; and *public policy* includes local, state, and national laws and policies.

Kaufman et al. (2014, p. 251) further adapted this SEM to analyse health behaviours. This model identifies five levels including *individual* (knowledge/information, risk perception, skills, disclosure, motivation, emotion, etc.); *interpersonal* (e.g. relationship power and equality, social support and trust, relationship satisfaction, communication level, etc.); *community* (stigma, peer pressure, norms, community organisation, cultural norms, etc.); *institutional* (health system, competent, supportive providers, friendly, culturally competent environment, support tools, etc.); and *structural* (poverty, access to services, cost of services, political context and priorities, funding for appropriate interventions, public policy, etc.).

The primary advantage of SEMs lies in their practical ability to identify and visualise the multiple nested layers of complex systems that contain the individuals and behaviours of interest. They show that behaviour must be analysed within a set of intertwined and overlapping systems that enable or constrain change. In this way, SEMs allow behavioural scientists to take into account complex systemic factors while being able to identify and establish relations among these determinants (McLeroy et al. 1988). However, significant challenges remain in achieving integration of social, ecological, and economic systems understanding in socio-ecological models. We demonstrate in the remainder of the paper how applying SEM and systems thinking to behaviour change addresses some of the most important ones.

2.3 Socio-ecological COM-B (SeCOM-B): incorporating the COM-B model into SEM

Individuals and systems are in a dialectical relationship in which change can begin from either direction (Hallsworth 2023). In our experience as applied behavioural and social scientists, organisational or government commissioners of behaviour change policies or programmes are typically focussed on creating measurable individual behaviour change within constrained resources and time. For that reason, enriching an individual-focussed model of behaviour change with a socio-ecological model that considers systems and structures requires an explicit, practical, and accessible logic for how structural factors interact with individual factors and eventually influence behaviour. The COM-B model, widely used in individual behaviour change interventions, has the advantage of presenting a framework with a ready-made connection to system-level influences through the *opportunity* component, which emphasises the role of the social and physical environment.

In response to the need to address "wicked problems" in sustainable development, and consistent with a recent call to synthesise knowledge systems developed by different disciplines and communities of practice (Moallemi et al. 2025, p.2), we devised a combination of the COM-B model and the socio-ecological model. We refer to this combined model as the SeCOM-B model. We recognise that we are not the first to attempt to combine individual analysis and system analysis in analysing behaviour change (de Salas et al. 2022; Yin et al. 2022; Nguyen et al. 2022; Jungbluth et al. 2024). For instance, de Salas et al. (2022) developed a framework called "the Rainbow Model of Ocean Health", which aims at tackling "wicked problems". This model places the individual within five other levels including community, technology, information and knowledge, policy, and international, national, and state. They also state that at the individual level, there is a need to connect individual behaviours with the COM-B model and the Behaviour Change Wheel to help link drivers of behaviours with intervention functions and policy categories (Michie et al. 2011). Another example is the multi-level framework developed by Boulet et al. (2021, 2022). This model includes micro (individual), meso (household), and macro (external to household) drivers of consumer behaviours. Yet they did not specifically link systems analysis to the COM-B model at the individual level.

Our SeCOM-B model represents a further adaptation and iteration of these efforts, with the aim of practically integrating analysis of individual behaviours with a systems approach. SeCOM-B acknowledges the dialectical relationship between individuals' behaviours and the systems and structures they are embedded within. Individuals, through their behaviours, maintain, reinforce, or even change the system structures, while simultaneously being enabled or constrained by the same systemic arrangements or structures. In developing this integrated approach, we also draw on the concepts of "crystallisation" (Ellingson 2009, p. 5) and "co-creation" (Frantzeskaki et al. 2025). Together, these concepts provide mechanisms and imply the use of multiple,

mixed-methods approaches through which social and behavioural scientists, communities, and other stakeholders can overcome differences and conflicts by building meaningful relationships, co-constructing a shared understanding of the challenge, co-envisioning a collective future vision, and co-developing practical, actionable solutions (Mertens 2015, p. 5; Frantzeskaki et al. 2025, p. 5).

The SeCOM-B model (Fig. 1) places a behaviour change intervention in a specific set of systems that enable or constrain the target behaviour under analysis. These enablements or constraints may be direct or indirect, manifest or latent, and may be expressed in the present or in the past (Kaufman et al. 2021; Yin et al. 2022; Nguyen et al. 2022; Jungbluth et al. 2024). In the SeCOM-B model, a behaviour change intervention for sustainable development can be defined fully only when it is considered in the context of the "wicked problem" of which it is a part, one characterised by complexity, uncertainty, manageability challenges, and interpretative opacity (Rotmans 2005, pp. 7–8).

Figure 1 illustrates how SeCOM-B helps design and facilitate a behaviour change intervention in the context of wicked problems. Its aim is to assist in changing a specific target behaviour over the shorter term (A)—*while taking account of its systemic context, causes and consequences*. This goal is often more suitable and feasible for the lifecycle of a typical behaviour change project. Target systems change (B) can be planned for, but this often needs a longer term to achieve.

Consistent with the COM-B model proposed by Michie et al. (2011, 2014), we specify that the target behaviour change is caused by a combination of an individual's *capability* and *motivation* to enact the behaviour, and the structural *opportunities* provided by their systems. A behaviour analysis (C) therefore aims to identify a specific primary and/or secondary target group whose behaviour needs to be changed, and to evaluate the individual factors (capability, motivation) that act as drivers or barriers to this change.

A systems analysis (D) enables the identification of structural opportunities for the behaviour change to occur. In the SeCOM-B model, the concept of "opportunity" in the COM-B model is expanded to include different structural facilitators and barriers that are embedded in multi-level systems in which the target behaviour is nested.

Based on Bronfenbrenner (1994, pp. 39–40; 1977, pp. 514–515), we consider the following nested systems:

- Microsystem (D1) refers to any immediate settings that contain and directly influence the individual and their target behaviour. Examples of major microsystems include school, home, and organisations. A microsystem also includes social groups whose critical relationships with the primary target group can influence the target behaviour.
- Mesosystem (D2) refers to any intermediate settings that contain the individuals' major microsystems and their interactions, which directly or indirectly influence the individual and their behaviour. The mesosystem can be adapted to include more than one systemic layer if needed. Examples of mesosystems are communities, economic sectors, and cities, which contain social norms,

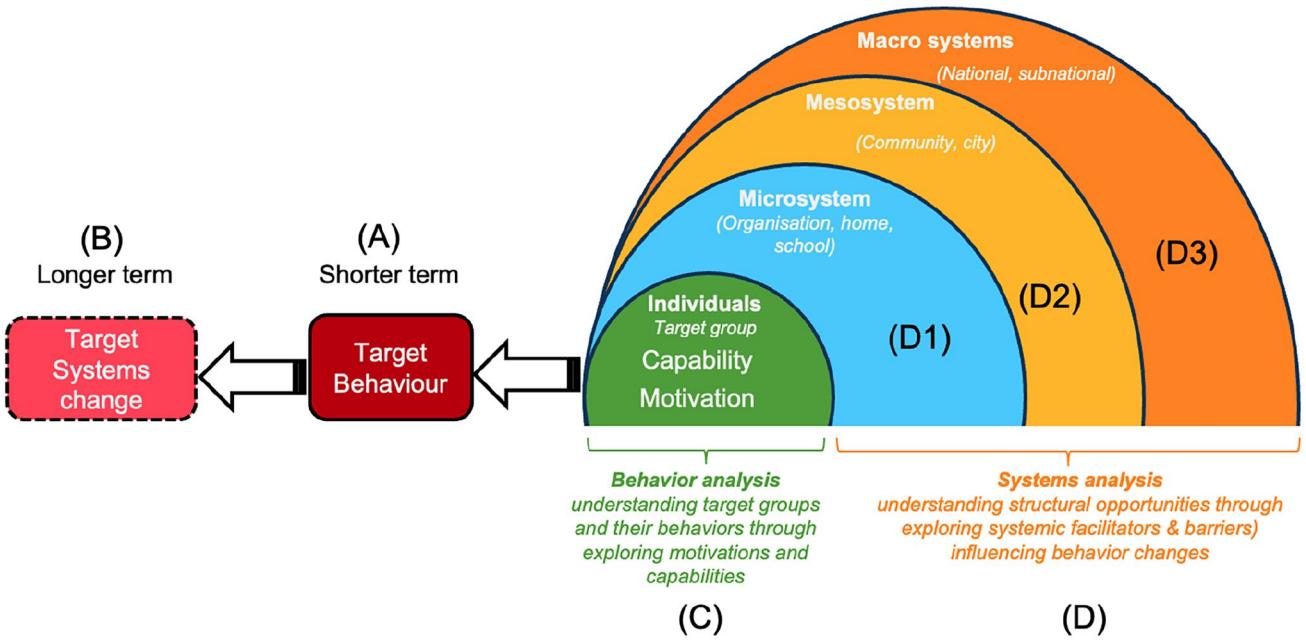


Fig. 1 A general socio-ecological COM-B (SeCOM-B) model

discrimination, social networks, industry practices, and standards.

- Macrosystem (D3) refers to the overarching institutional structures of a given culture or subculture, including economic, social, educational, legal, and political systems. These structures both shape and are reflected in the micro- and mesosystems, acting as channels that, explicitly and implicitly transmit ideology, meaning, and information to individuals, social roles, and organisations (Bronfenbrenner 1977, p. 515).

Explicitly recognising the existence of, and interactions between, multiple, theory-informed layers of influence significantly expands the applicability of the COM-B model. These nested systems are interdependent through the process of forming and implementing social structures (Bronfenbrenner 1977, p. 518). This is because "systems can be nested within systems" (Meadows 2008, p. 15), and structures can be nested within structures. In other words, structures of higher-level systems can direct and at the same time are influenced by the formation and execution of the structures of their lower-level systems. For instance, national policies on green infrastructure can direct the development of a city's policies on nature-based solutions, or local community norms on recycling can be influenced by individual households. The next section explores the application of the SeCOM-B model in three case studies of behaviour change interventions (see Table 1).

3 Our methodological approach, context and materials

3.1 Contexts and materials

In this study, our research team includes five researchers who applied the SeCOM-B model in behaviour change projects from 2022 to 2023. From the projects in which we applied this model, we chose three case studies (Table 1) that were behavioural science-driven projects seeking to study behavioural factors and build behavioural interventions to address a specific problem within the broad category of sustainable development. These three cases represent diverse socio-ecological systems and "wicked problems": gender inequality in the context of Vietnam, a developing country in the Southeast Asia region (Case 1), farmers' participation in carbon markets in three states of Australia (Case 2), and Dja Dja Wurrung Traditional Owners' climate change adaptation in local councils in the state of Victoria, Australia (Case 3). This selection demonstrates the broad relevance and applicability of SeCOM-B. Below we briefly describe the context and methods used in each case study.

Case 1 focussed on developing a behaviour change communication strategy for promoting gender equality in Northwest Vietnam to support the transition of the Gender Responsive Equitable Agriculture and Tourism (GREAT) programme from Phase 1 (2017–2022) to Phase 2 (2022–2027) (Nguyen-Trung et al. 2022). The programme, funded by the Australian Department of Foreign Affairs and Trade (DFAT), sought to empower ethnic minority women in the Son La and Lao Cai provinces in agriculture and tourism market systems (Aus4Equality 2024). In this project, the target group was ethnic minority women in northern Vietnam, and the target behaviour was broadly conceptualised as confidently participating in community leadership (e.g. running for leadership positions, voicing their ideas to influence the community's change). This group and their behaviours were situated in Vietnamese society, organised in a multi-level governance structure: central governance, provincial governance, district governance, community governance, and village (community-based unit). According to the programme documents, structural inequalities and power imbalances were evident not only at the household/family level (e.g. husbands often played the role of household heads), but also at the community level (e.g. women rarely participated in the leadership structure of their village), the government level (e.g. women were absent from the government's decision-making processes), and the cultural level (e.g. social norms favouring masculine dominance). These structural gaps maintain gender discrimination and stereotypes that prevent ethnic minority women from playing more powerful roles in their community leadership. This background presented a "wicked problem" that excluded women from gaining access to political capital and decision-making processes.

The client had already used an adapted version of the socio-ecological model to map their theory of change against the multi-level governance structure (GREAT 2022). In response to their request to add a behaviour change aspect, we combined the COM-B model and a socio-ecological model to create an adapted SeCOM-B model, which was used to map multi-level influences on women's behaviours. We determined that the individual (i.e. women) was the main unit of analysis. This means that these women were the main focus of our behaviour change intervention. Their individual factors (capability, motivation) were situated within nested systems (representing "Opportunity" in the original COM-B model) that included the microsystem (i.e. family, interpersonal relationships with husbands), the mesosystem (i.e. local community at the village level including gender norms, village's stigma, and the agricultural and tourism sectors' networks, institutions such as businesses and service providers), and the macrosystem (i.e. Vietnam's public policy and regulations concerning gender equality, agricultural and tourism development, and funding opportunities).

Table 1 Overview of SeCOM-B case studies

Aspects	Case 1	Case 2	Case 3
Problem	Gender inequality and stereotypes prevent ethnic minority women from participating in community leadership	Farmers seeking to transition from extractive agriculture are reluctant to participate in environmental markets that may facilitate this	Traditional Owners are left out of local councils' climate adaptation planning
Goal	To develop a behaviour change communication strategy to change gender norms in Northern Vietnam	To evaluate how well a tailored intervention addresses the barriers and solutions associated with farmers' participation in environmental markets	To co-design an intervention that helps local councils incorporate Dja Dja Wurrung people's knowledge into their climate adaptation planning
Broad target behaviour	Ethnic minority women's participation in community leadership	Farmers' preparation for, participation in, and shaping of environmental markets	Local councils and their officers incorporating Traditional Owners' knowledge into their climate adaptation planning
Unit of analysis	Individual (individual ethnic minority women)	Individual (individual farmers) participating in landscape-scale collective projects, facilitated by intermediary organisations	Local councils as organisations (main unit of analysis) Individual (council officers as sub-unit of analysis)
Microsystem	Family, including women's interpersonal relationships with husbands	Farmers' farm business, goals and capabilities, and the influence of interpersonal networks including close friends, family Intermediary organisations (e.g. roles of cooperative catalyst groups)	Organisation (local councils including their resources, staff, and relationships with DJAARA communities)
Mesosystem	Local community or village (e.g. gender norms); Economic sector including the agricultural and tourism sectors' institutions	Farming community including traditions, stigma, and social norms Economic sector including agricultural institutions, input providers, and technologies	Community including DJAARA Traditional Owner communities' traditions, holistic approach to nature-people relationship
Macrosystem	National structures including public policy, laws, and funding programmes	Nation/state policies, historical legacies, biodiversity, and policy/laws on environmental markets	Nation and state biodiversity, climate change, and disaster management policies, and infrastructure systems Historical legacy of conflict and oppression by government towards Traditional Owners

Source: Authors

We applied the SeCOM-B framework through a multi-method approach to collect and analyse data. This included a literature review conducted from 5 April to 15 May 2022, which aimed to identify key behaviour change and behaviour change interventions targeting gender norms and women's economic empowerment in key sectors in accordance with the layers of the SeCOM-B model. The review method included a grey literature review, drawing on a significant volume of evidence and data which the GREAT programme team had collected, either directly, or through multiple evaluations (e.g. Design Update for GREAT-2, Mid Term Review), and a rapid evidence review using SCOPUS, Google Scholar, and the Vietnam Journals Online (VJOL) databases. This method yielded 1,144 references, screened 1082 studies after removing duplicates, to identify 13 primary studies and 2 reviews for data extraction. The review method was followed by qualitative interviews conducted from 17 to 31 May, 2022. This method included four consultation workshops and two key informant interviews. The workshops included one workshop with GREAT's Project Management Units (PMU) in Lao Cai and Son La provinces of Vietnam, two workshops with provincial and district-level stakeholders in these two provinces (such as the Department of Education and Training (DOET), the Department of Labour, Invalids, and Social Affairs (DOLISA), the Women's Union), and one workshop with the GREAT team in Hanoi and DFAT's representatives. Two key informant interviews collected data from the programme's communication and gender specialists.

In Cases 2 and 3, we draw on a programme in Australia called the Climate Adaptation Mission (Saeri et al. 2023). In Case 2, we aimed to understand the barriers and drivers associated with farmers' participation in environmental markets (e.g. carbon and emerging biodiversity markets) in three regional areas: the Traprock region Landscape Impact Accelerator programme (Queensland, Australia), the Goulburn Broken region Carbon + Biodiversity Program (Victoria, Australia), and the Environmental Markets Leadership Program (New South Wales, Australia) (BehaviourWork Australia 2023a). A prior literature review highlighted the multi-level nature of influences on behaviours relevant to environmental markets (Kaufman and Meis-Harris 2022). In Case 2, the unit of analysis (and target group) was individual farmers, as well as local organisations with the purpose of connecting farmers to environmental markets. By conducting interviews with farmers, their local networks, and the local organisations and in a subsequent literature review, we found that there were several distinct influences on farmers' decisions to engage in environmental markets. These were in part a cognitive analysis of the economic cost and benefit of participating, or judgements of the market's immaturity. Crucially, however, farmers were also influenced by their judgement of anticipated acceptance by peer groups if they participated in the programmes, the support by local organisations, potential impacts on the local community, the availability of natural capital in their region, the agricultural sector, and the national and state policies towards the farming community (Kaufman & Nguyen-Trung 2023). These multi-level and multi-actor layers demonstrate that encouraging participation in environmental markets is not a simple behaviour change issue but is a "wicked problem", and requires both a socio-ecological analysis and a model to understand how these structural factors interact and influence farmers' behaviours. These insights helped identify the crucial intermediary role played by local organisations and the intervention programme in linking individual and system influences.

To examine these multi-layered influences, we employed a mixed-methods approach tailored to each regional context. Between October 2022 and June 2023, we conducted pre- and post-programme surveys, semi-structured interviews, and observational participation in programme delivery. In the Traprock region (Queensland), the Landscape Impact Accelerator programme was evaluated through surveys with pilot farmers ($n = 16$), followed by semi-structured interviews with both farmers ($n = 5$) and the expert team involved in implementation ($n = 8$). In Victoria's Goulburn Broken region, we surveyed participating farmers ($n = 16$) before and after their involvement in the Carbon + Biodiversity Program. We also conducted interviews with farmers as well as with representatives from catchment management groups, consultants, NGOs, and local councils. In New South Wales, where the Environmental Markets Leadership Program enrolled 87 farmers, a pre-programme survey was completed, and semi-structured interviews were undertaken with key stakeholders such as government agencies, industry bodies, and natural resource management (NRM) organisations. These data collection activities allowed us to triangulate farmer perspectives with those of programme implementers and support networks, providing a nuanced understanding of behavioural drivers, local enabling conditions, and institutional dynamics influencing farmers' engagement in environmental markets.

Case 3 was a separate stream of work in the Climate Adaptation Mission (Saeri et al. 2023). Here, we collaborated with our partners from the Dja Dja Wurrung Clans Aboriginal Corporation (DJAARA), 12 local governments (councils) in Victoria, state government bodies, and other climate groups to co-design and co-organise a Dja Dja Wurrung-led Climate Adaptation Forum (BehaviourWorks Australia 2023b). In this Forum, DJAARA invited participants from Victorian local governments to learn about their values, climate strategy, and plan for healing Country ("*Dhelkunya Dja*"). In this context, our goal was to

assess whether the intervention could help local councils incorporate Dja Dja Wurrung people's knowledge into their climate adaptation planning. The exclusion of Dja Dja Wurrung culture and knowledge from local governance processes signifies not only the lack of capacity at the individual level (i.e. local officers) but also implies local governance's lack of in-place resources, leadership, and innovation culture, and missing social capital between local councils and Dja Dja Wurrung people. These symptoms again suggest the need to apply a multi-level framework to diagnose and design a relevant behaviour change intervention. To research this goal, the main unit of analysis was the local governments. Officers from local governments who participated in the Adaptation Forum were considered as the *sub-unit of analysis*. In short, the target of interventions was not only the change in behaviours at the individual level, but also the change in organisations at the microsystem level.

In Case 3, we employed several methods: (1) Background research from 25 to 30 June 2022, which focussed on understanding the system and context of 12 councils in Victoria, Australia. We conducted secondary data analysis of 12 local governments' records and documents to understand their climate change strategy, governance structure and resources, and their relationship with DJAARA. We also conducted key informant interviews ($n = 3$) with local government officers to understand their individual perspectives on drivers and barriers to the target behaviour. (2) The pre-Forum assessment was conducted in November 2022. It involved a quantitative survey of 12 staff members from eight local governments in the study setting, and qualitative interviews with seven staff members from seven local governments in the study setting. This assessment helped confirm the findings from the background research and provided a benchmark for the post-Forum assessment. (3) The Adaptation Forum was held on 30 November 2022 at No. 7 Park, outside Bendigo, on Dja Dja Wurrung Country. This Forum was organised as an interactive, participatory focus group that involved discussions with participants on the factors that prevent or support the incorporation of Dja Dja Wurrung people's knowledge into the local governance's climate adaptation planning. The post-Forum assessment was conducted in April and May 2023. It involved a quantitative survey of six staff members from five local governments in the study setting, and qualitative interviews with four staff members from local governments, as well as three other participants in the Adaptation Forum including representatives from DJAARA, the Victorian Department of Energy, Environment, and Climate Action (DEECA), and the Central Victorian Greenhouse Alliance (CVGA). Following the formal evaluation activities, we conducted an online scale-up workshop with 11 participants on 5 July 2023 focussed on the behaviour change intervention. The intent of the workshop was to reflect on key lessons from designing, delivering, and/or participating in the intervention; situate the intervention in the context of other climate adaptation activities and interactions between local government and DJAARA; and increase the reach and impact of the behaviour change intervention.

3.2 Critical reflection as a research method

In this study, we did not aim to reanalyse the primary data collected in each case study. Instead, based on our previous research (Nguyen-Trung et al. 2025), we employed critical reflection to help us critically reflect on our application of the SeCOM-B model in our behaviour change projects that aimed to address "wicked problems" in sustainable development. Critical reflection should be distinguished from reflexive practices, which are part of a research process where researchers reflect on their biases and how they affect their research actions (Braun and Clarke 2019a, b). Critical reflection is a distinct method of inquiry (Fook and Gardner 2007, 2012) that can help research projects overcome the challenges driven by systemic complexity, interdisciplinary and transdisciplinary partnerships, and stakeholder engagement (Black et al. 2023). The data created by this method include individual reflections, collective reflections, and notes from group discussions, which can distil lessons learnt in the research project cycle (Nguyen-Trung et al 2025; Fook 2012).

The lead author was involved in all three case studies, while the second and fifth authors were involved in the second and the third case. The third and fourth authors did not directly participate in the three projects, but they synthesised behavioural and systemic analyses at different levels and offered their crucial reflection on the application of SeCOM-B based on other projects. They were invited to add a more critical perspective to the authors involved in these three projects. Formed in July 2023, the team then conducted a series of critical reflection sessions to collect data from team members until May 2024.

In the process of critically documenting our application of the SeCOM-B model, we started our critical reflection sessions by sharing a shared Google Docs, which outlined key questions such as:

- What are the advantages and disadvantages of the COM-B model in addressing sustainability challenges?
- What are the advantages and disadvantages of SEM models in addressing sustainability challenges?
- How can SeCOM-B overcome these challenges in analysing barriers and drivers of sustainability problems?

- How was SeCOM-B applied in the three case studies? What are the limitations of the SeCOM-B model? How can we improve its applications?

The research team then reflected on their experiences with regard to these questions and wrote their personal reflections. The team then reflected on others' reflections and asked follow-up questions. For instance, one team member wrote that the advantages of SeCOM-B models were that they had the capacity to lay out different systemic levels so that we could identify the structural forces that influence a sustainability problem. This prompted a follow-up question asking how to address the links between different systemic layers and how to translate these forces to create behaviour change at the individual level. By creating a critical dialogue, we were able to investigate the promise and limitations of applying the SeCOM-B model. The next section presents our reflection on the application of SeCOM-B in three case studies.

4 The applications of SeCOM-B in three case studies

4.1 Case 1: Communication strategy for gender equality behaviour change in Northern Vietnam

Figure 2 illustrates Case 1, in which we employed the SeCOM-B model to analyse the barriers and facilitators that affect the active participation of ethnic minority women in community leadership in the Northern mountainous areas of Vietnam (Nguyen-Trung et al. 2022). The COM-B model assists in categorising the personal factors that influence or restrict women's active participation in acquiring leadership positions in public spaces (e.g. village heads). When we applied this model to co-analyse the drivers of behaviours with diverse stakeholders unfamiliar with behavioural science, we found that their experience was positive as the model helped them quickly understand the various influences contributing to the target behaviours.

Based on the general SeCOM-B model (Fig. 1), we adapted the systemic layers and structural factors at each level to make them relevant to the contextual conditions of Vietnamese society and gender inequality (Fig. 3). In response to the request of our client, who aimed to target their behaviour change communication campaigns not only at women but also at other influential groups, we focussed on mapping out key factors and key groups at each level that played a crucial role in influencing the target behaviours of women.

- *Individual level:* Ethnic minority women, as the primary target group, are placed at the centre. At this level, factors such as their knowledge, information, risk perception, motivation, emotions, and skills are considered crucial barriers to their desired behaviours in business development and participation in local politics.

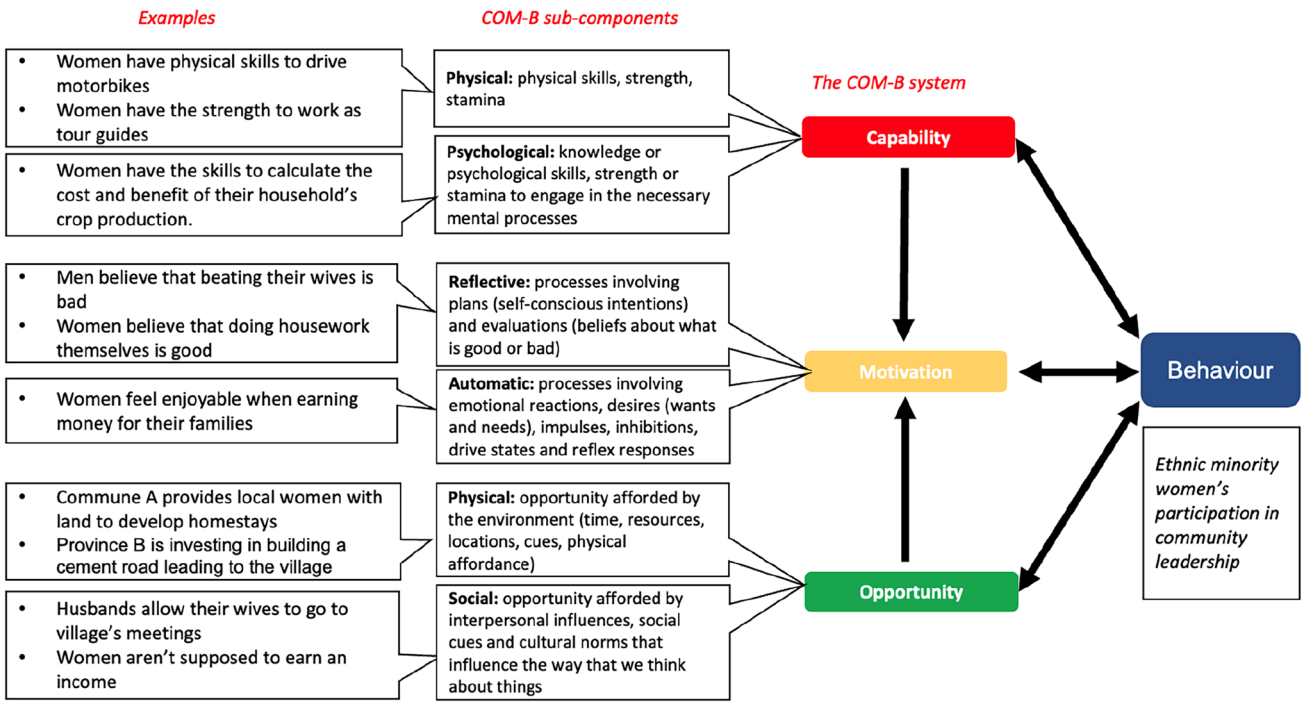


Fig. 2 COM-B model and examples in a gender-related intervention project

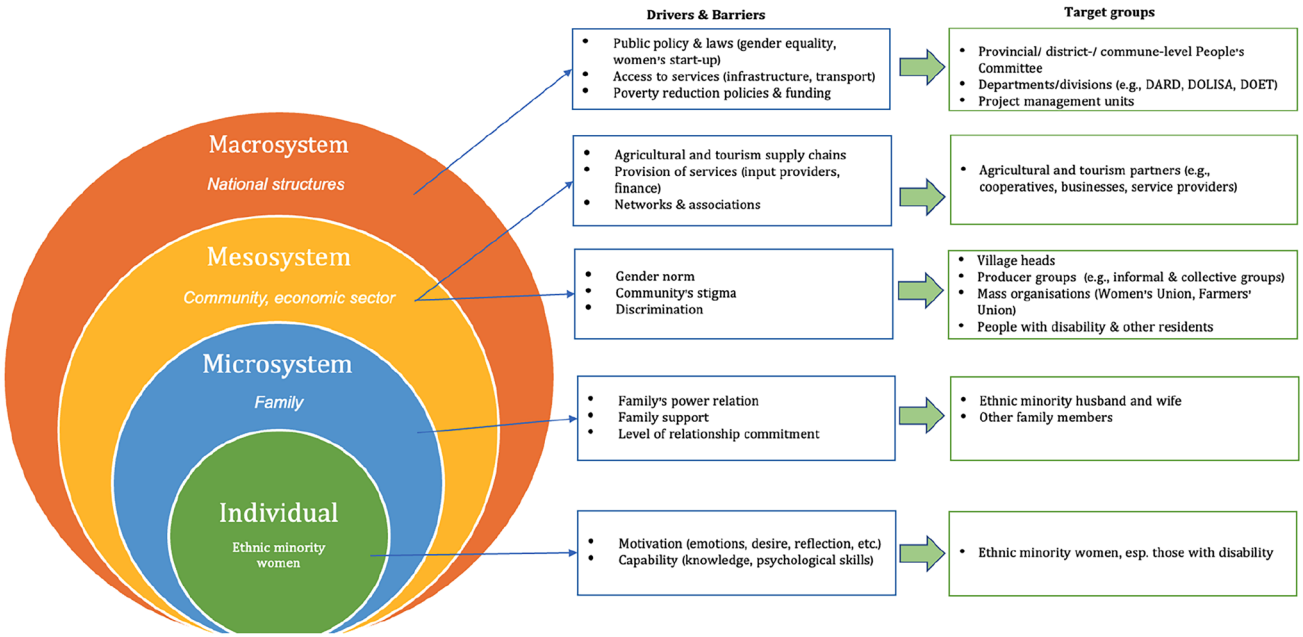


Fig. 3 Case 1. Adapted SeCOM-B for a gender-related behaviour change project

- *Microsystem level*: This level focuses on women's immediate relationships, especially the family's power and equity, support and trust, relationship satisfaction, communication level, relationship health, and level of commitment. These factors are considered in relation to women and their close networks, such as husbands and other family members in the extended family (e.g. relatives).
- *Mesosystem level*: This level includes the *local community's* factors such as stigma towards women, peer pressure, gender norms, community organisation, cultural norms, and the positions of religious, cultural, and opinion leaders that could affect the advancement of women's economic empowerment. It also covers *the agriculture and tourism sectors*, which reflect institutional arrangements including supply chains, the growth of service provision (including agriculture input providers, tourism information providers, tourist guides, etc.), and the networks of these businesses.
- *Macrosystem level*: This level considers Vietnam's national and provincial public policies and laws regulating gender equality, funding programmes for interventions associated with gender equality and poverty reduction in ethnic minority areas, access to services, transportation infrastructure, and disaster risk management and climate change adaptation policies and strategies.

Based on discussions with our client, we eventually focussed our intervention on the first three levels (individual, microsystem, and mesosystem) while referring to the macrosystem's factors where appropriate. This emphasis reflected anticipated budget limitations and time constraints. The application of COM-B allowed us to identify key secondary target groups for the behavioural intervention (in this case, a behaviour change communication intervention). For instance, at the microsystem level, it was necessary to involve husbands, as their motivation, capability, and behaviours regarding women's participation in public leadership represented significant opportunity factors for women's desired behaviours. At the mesosystem level (i.e. the community), we identified village heads whose capacity and motivation acted as influential factors enabling or constraining women's behaviours in community spaces. Figure 4 demonstrates that to change women's behaviours (i.e. encourage them to actively run for and hold community leadership positions), a behaviour change communication campaign must also foster changes in the behaviour, capability, and motivation of husbands and village heads.

4.2 Case 2: Australian farmers' participation in environmental markets

In Case 2 (Fig. 5), to analyse the barriers associated with farmers' participation in environmental markets, we applied an adapted version of the SeCOM-B model, placing farmers as the main unit of analysis (Kaufman & Nguyen-Trung 2023). As can be seen, capacity and motivation are considered at the individual level, which represents the individual

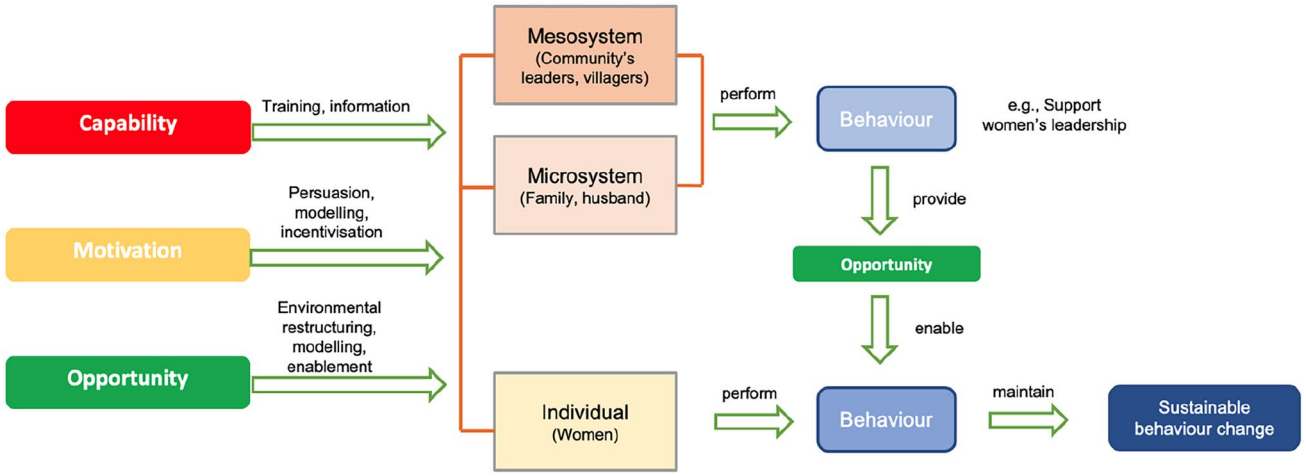


Fig. 4 Case 1. Designing behaviour change interventions in the SeCOM-B model

farmers situated within nested collective or institutional layers. The opportunity component includes the following socio-ecological levels:

- *Microsystem level:* This level refers to farmers' immediate settings and close relationships such as friends, family, relationship strength, social support, information exchange, and trust. It includes the farm business, and the farmer's goals and capabilities for it.
- *Mesosystem level:* This level includes factors associated with *farming communities,* with the presence of stigma, tradition, social norms, social acceptance, social groups,

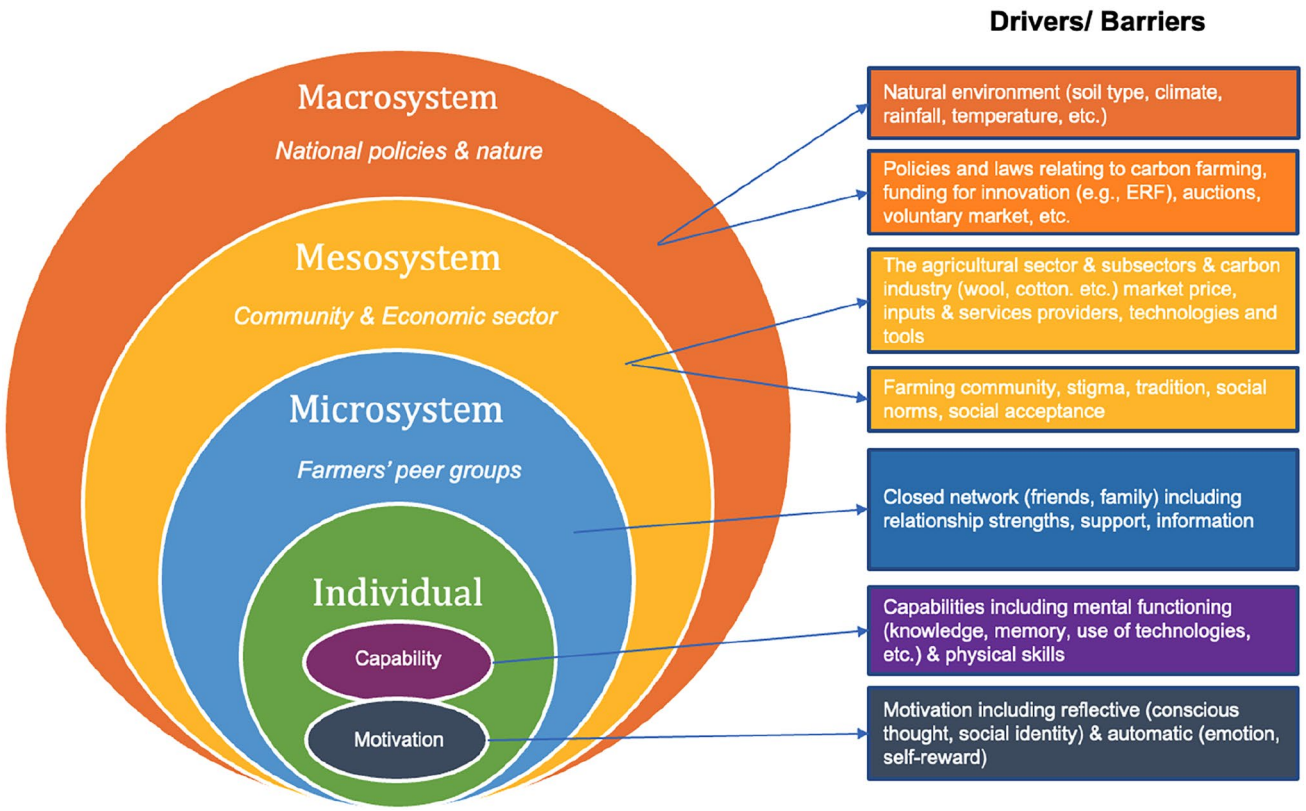


Fig. 5 Case 2. Mapping in farmers' participation in environmental markets

local land care networks, and local advisors, which influence farmers' decisions to join environmental markets. It also includes factors associated with the economic sector of agriculture including cropping, dairy, meat, the carbon industry, market mechanisms, suppliers, sellers, and financial actors.

- *Macrosytem level:* This level includes factors associated with national or state legal systems that are relevant to the establishment and operation of environmental markets, the state's funding for carbon projects (such as the presence of the Emission Reduction Fund), and voluntary markets policies. It also covers factors associated with the state's natural environment including the presence of natural capital (soil, rainfall, land, climate, temperature), which are of crucial importance in farmers' assessment of their opportunities to join carbon projects.

All of these systemic layers and factors are considered significant to farmers' decision-making and participation in environmental markets.

4.3 Case 3: Analysing local councils and their officers' behaviours in climate change adaptation involving Traditional Owners communities

In Case 3 (Fig. 6), the project targeted the intervention at both the individual and microsystem levels. In other words, the SeCOM-B model was adapted to involve local council officers (individuals) within their organisation, that is, the microsystem. In this project, an adapted SeCOM-B was developed as follows:

- *Individual level:* This level explores the capabilities of local council officers, including their cultural awareness of Dja Dja Wurrung people and culture, their confidence and motivation to incorporate Traditional Owners' knowledge into their council's work, and their capabilities to undertake follow-up activities after the Adaptation Workshop.
- *Microsystem level*: This level explores local councils' leadership, organisational culture, innovation, governance, and decision-making, which influence council officers' capability, motivation, and behaviour in working with the Dja Dja Wurrung Clans Aboriginal Corporation and communities.
- *Mesosystem level*: This level explores factors associated with the regional Traditional Owner communities, including the Dja Dja Wurrung community as the main focus of data collection, and other Traditional Owners communities whose land is governed by local councils.
- *Macrosystem level:* This level explores factors associated with Australia and the Victoria state's natural resources, climate-induced risks, local natural resources, biodiversity, and the built environment (e.g. infrastructure). While these levels are not exhaustive, they represent the key findings of our intervention evaluation (Saeri et al. 2023).

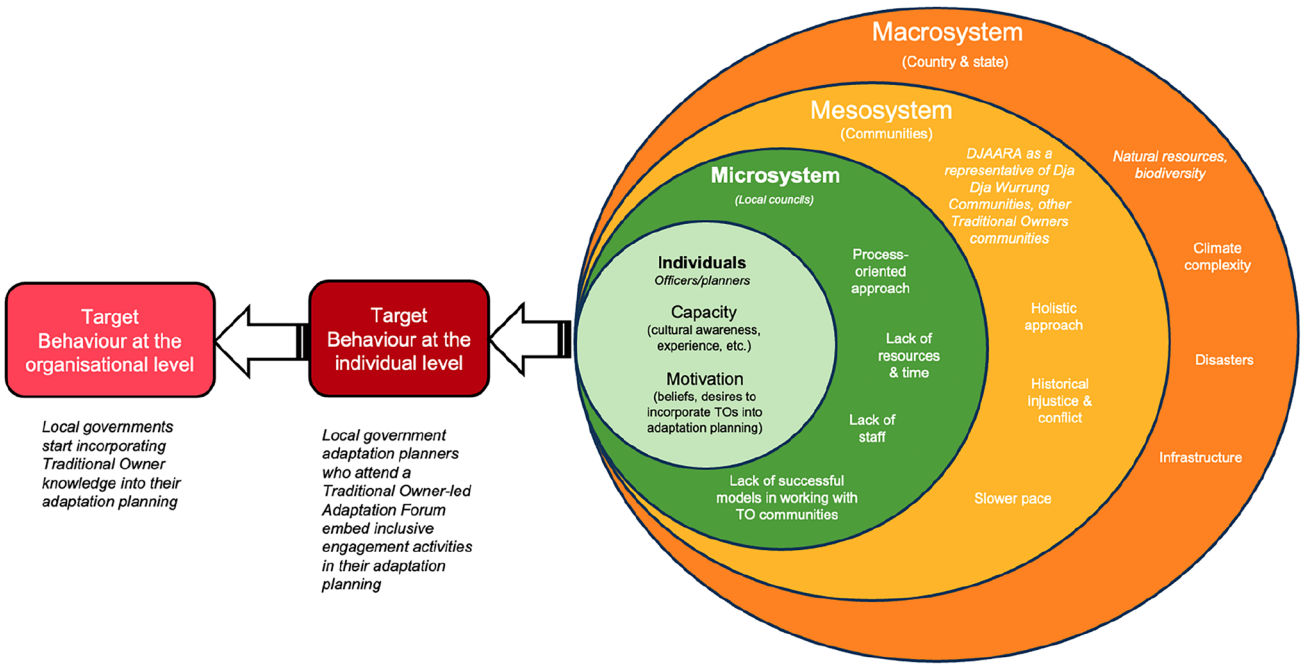


Fig. 6 Case 3. SeCOM-B in climate adaptation planning project

5 Discussion: implications for applying SeCOM-B in projects

In the last 5 years, there has been a rise in systematic reviews that apply a combination of behaviour models and systems approaches (Yin et al. 2022; Nguyen et al. 2022; Macklin et al. 2023; Jungbluth et al. 2024). These reviews were conducted in different sectors of behaviour change: determinants of children and urban green infrastructure (Yin et al. 2022); factors influencing alternative protein consumption (Nguyen et al. 2022); drivers of household recycling behaviours (Macklin et al. 2023); and factors affecting hospital recycling behaviours (Jungbluth et al. 2024). For instance, Jungbluth et al. (2024, pp. 3–4) conducted a mixed-methods systematic review, screening 9188 papers and extracting data from 27 studies. To map the influences on hospital recycling behaviour, they combined the COM-B model with a multi-level framework. They found that influences on hospital recycling behaviour operate across multiple levels. At the micro-level, individual factors such as concerns about infection control, lack of knowledge, pro-environmental considerations, personal recycling habits at home, and individual interest in recycling shape staff behaviour. At the meso-level, influences are rooted in the hospital's internal environment, including lack of access to recycling bins, insufficient staff to dismantle recyclables, unsupportive colleagues, absence of leadership support, limited time and training, and a lack of internal policies or incentives for recycling. At the macro-level, broader contextual factors come into play, such as product design preventing recycling, the availability and effectiveness of external waste management systems, national or regional regulations, and the cost implications of waste treatment if not recycled. The studies suggest that the combination of COM-B and multi-level models helps add nuance to both individual/behaviour analysis and system analysis.

Based on the rigour of systematic reviews, we argue that integrated individual and systems models demonstrate the advantages for unpacking "wicked problems" by identifying and conceptualising the drivers and barriers of behaviour change. They imply the significance and applicability of combined models in different contexts. These studies, while demonstrating the capacity of individual-system integrated models, were limited to the literature review scope. In this regard, they did not offer practical guidance on how to apply integrated frameworks when setting up a real-world project and designing a behaviour change intervention. This is the gap our study addresses. In this study, we present three case studies in which an integrated socio-ecological and COM-B framework (SeCOM-B) was used to help practitioners, including researchers and project staff, to understand the drivers and barriers of wicked problems, as well as develop specific designs of behaviour change interventions. Based on our experience and lessons learnt, we outline below some theoretical and practical considerations that need to be taken into account when applying the SeCOM-B model to a specific behaviour change intervention.

5.1 Conducting an initial SeCOM-B mapping

Traditional behavioural science has been largely concerned with the individual. Recent initiatives in behavioural public policy and applied behavioural science, on the other hand, require us to broaden our approach beyond individual analysis to account for the context in which behaviours occur (Hallsworth 2023). But which social contexts should we consider? There are many interconnected levels of social systems. This requires a systems analysis, examining what Chater and Loewenstein (2022, p. 1) refer to as the "s-frame", which is more concerned with systems of rules, norms, and institutional arrangements.

In this sense, the SeCOM-B model can account for the complexity of nested systems through a SeCOM-B mapping that displays complex nested layers of systems surrounding the target behaviour. A SeCOM-B mapping, different from a standard system mapping, includes both COM-B and SEM models, with the goal of mapping the individual and their behaviours in a unique collection of nested systems (see Figs. 3, 5, and 6 for examples). The prominent strength of this mapping framework is that it can facilitate integrated thinking that moves beyond the extremes of either individual-focussed or systems-focussed approaches to combine different scales of analysis in behaviour diagnosis, especially when analysing the target behaviour's drivers and barriers (Kaufman et al. 2021). By doing so, the exercise assists researchers and stakeholders in identifying different systemic layers and structural forces that are crucial in shaping the target behaviour. The socio-ecological layers within each of the nested systems (microsystem, mesosystem, and macrosystem) should be chosen, altered, and labelled to represent the unique context and characteristics of specific behavioural issues of interest. Additionally, by retaining the elements of the COM-B model, this mapping allows researchers to focus their interventions on behaviours as the main unit of analysis/intervention while acknowledging their interdependence with the structures of nested systems (Kaufman et al. 2021; de Salas et al. 2022). We argue that integrating COM-B and SEM approaches presents behavioural systems as a useful "boundary object"—providing a point of contact between different disciplinary and practice understandings of problems and possible solutions through the lens of behaviour. Our case study (Table 1) shows three possible ways of conceptualising and labelling the socio-ecological system levels based on the central problem, the chosen context, target group, and target behaviour under study. Following a SeCOM-B mapping, researchers can identify, prioritise, and link barriers at individual and system

levels to the actors responsible for making the necessary changes (either at behaviour or system levels).

5.2 Identifying the unit of analysis and drawing the boundaries

Identifying the unit of analysis (or unit of intervention) is critical because it helps determine the focus of the intervention and/or research, and guides decisions about who, what, where, when, and how to gather data. A unit of analysis is the case of the intervention, which is defined as a "bounded system", or a system with a clearly defined boundary (Merriam and Tisdell 2016, p. 99). Drawing a system's boundaries helps distinguish the case(s) (unit of analysis) from their socio-economic and natural contexts (nested systemic layers). An intervention can include multiple primary units of analysis (i.e. multiple cases) or a single case with multiple embedded sub-units of analysis (Yin 2018, p. 87). For instance, one intervention might compare City A and City B; another might focus on City A, treating its organisations as sub-units. Time, place, context, and social group are important criteria to consider when establishing boundaries around units of analysis (Yin 2018, p. 65; Frantzeskaki et al. 2025, p. 5), but researchers can also choose others such as the type of climate-induced impacts, disaster events, or administration level.

Individuals and their behaviours are often the unit of analysis because behaviour change interventions focus on them. Yet, in certain circumstances where system change is targeted, researchers may consider higher system levels such as social groups, households, organisations, or communities as the unit of study (see Fig. 6: Case 3). Extending the goal from behaviour change to system change often requires a longer timeframe to address (see Sect. 2.3) (Pereira et al. 2020, p. 172). Depending on the study's unit of analysis, the choice of systemic levels can vary. For example, if we view the organisation (a microsystem) as the main unit of analysis, everything outside organisations constitutes the systemic context. Individuals within the organisation can be considered a sub-unit of analysis in this scenario. Other aspects of an organisation, such as a department, can be likewise viewed as a sub-unit of analysis.

5.3 Connecting SeCOM-B levels with behaviour interventions

What makes the SeCOM-B model different from traditional socio-ecological models (SEMs) is its emphasis on behaviour change interventions. The SeCOM-B model keeps the COM-B components (capability, motivation, behaviour) at the centre and links them to the Behaviour Change Wheel or systemic approaches (Moallemi et al. 2025). According to the Behaviour Change Wheel (Michie et al. 2011, p. 6), there are two levels of creating behaviour change once barriers have been examined: First are *intervention functions*, which include nine approaches such as education, persuasion, incentivisation, coercion, training, restriction, environmental restructuring, modelling, and enablement. Second are *policy categories*, which include communication/marketing, guidelines, fiscal, regulation, legislation, environmental/social planning, and service provision. Policies enable or support intervention functions, which in turn influence behaviour (Michie et al. 2011, p. 1). This suggests that there are two pathways for SeCOM-B design: behaviour change interventions (discussed in this section) and system change interventions (discussed in Sect. 5.4).

Behaviour change interventions often include education, persuasion, incentivisation, coercion, training, environmental restructuring, modelling, and enablement (Michie et al. 2011, p. 8). These interventions focus on the entry points that are changes in the *capability, motivation,* or *behaviours* of the target individuals located at each system level (e.g. women as the main target group, husbands at the microsystem level, village heads at the mesosystem level). SeCOM-B also encourages practitioners to identify secondary target groups: actors whose critical relationships with the primary target group might play different roles "in shaping desirable transitions and transformations through [their] transformative agency and governance" in the target system (Hölscher et al. 2018, p. 2). The capability, motivation, behaviours, and opportunity of these secondary groups often play an essential role in preventing (opportunity as structural barriers) or promoting (opportunity as structural facilitators) the main target group's behaviour change. In short, focussing on the primary and secondary target groups "emphasi[ses] creating the social attributes—capacities—that empower individuals and communities to take action on their own behalf" (Scoones et al. 2020, p. 67).

For instance, in Case 1 (Fig. 3), to initiate behaviour change in ethnic minority women, it was crucial to change key stakeholders in the community. It was found that one of the barriers was that male village heads lacked capability (specifically knowledge and specific skills in organising gender-inclusive activities). The proposed behaviour intervention therefore proposed training on gender-inclusive principles to raise awareness, offer knowledge, and build gender-sensitive skill sets among these leaders. Enhancing their capability, motivation, behaviour, and opportunity was expected to cascade to the main target group's (women) capability, motivation, and behaviours. The literature on gender equality confirms that "[changes] among men represent the next phase of the movement for women's equality" and "men must be involved in the transformation" (Kimmel 2009, pp. 359–360). Similarly, in Case 3, we found that council leaders at the microsystem level (local councils) held the authority to embed Traditional Owners' knowledge in

developing their councils' adaptation strategies. Behaviour change interventions targeting this group therefore became a key recommendation for future work. To summarise, intervention design in SeCOM-B should continue to use behaviour-focussed frameworks such as the Behaviour Change Wheel and COM-B model to connect actors at each level to appropriate behaviour change interventions.

5.4 Recognition of cross-cutting forces in system interventions

The second approach is system change interventions. This type's entry points are structural shifts (i.e. changes in rules, institutions, resources) that are typically embedded at the microsystem, mesosystem, and macrosystem levels. It is critical to recognise the interdependence between social structures across system levels (Bronfenbrenner (1994, pp. 39–40). It is often assumed that the structures of the higher-level system often have great impacts on the structures of the lower-level system.

In Case 1, the dominance of male chauvinism at the mesosystem level (i.e. village community) influences not only village heads' organisation of gender-sensitive activities at such a level, but also influences women as individuals and their relationships with their husbands (the microsystem level, or family). In this regard, Michie et al. (2011, p. 8) proposed several structural interventions including communication or marketing, guidelines, fiscal measures, regulation, legislation, environmental or social planning, and service provision. These interventions are designed to target changes at system levels, which in turn can create favourable conditions for behaviour change.

Similarly, examining the barriers and facilitators experienced by farmers participating in environmental markets (Case 2) highlights that policy and market design problems can be partially mediated by trusted organisations, but that ultimately, for this particularly pragmatic group, the final decision reflects "what works" in a delicately balanced socio-ecological system. Importantly, identifying a bottleneck of upfront costs for rigorous baseline ecosystem assessment meant that in the next implementation of the programme, the farmer-owned cooperative could reallocate resources to cover this, increasing uptake and comfort with the programme.

System transition and transformation approaches also provide SeCOM-B models with relevant interventions. These approaches aim to identify specific *entry points* to leverage structural changes across levels. For instance, Moallemi et al. (2025, p. 1) identified six entry points to understand, initiate, and sustain systemic change. There are three groups (Moallemi et al. 2025, pp. 4–6): Group 1 focussing on building momentum (mainstream emerging innovation, overcome incumbency, leverage positive pressures); Group 2 focussing on navigating pathways to future states (account for multisector dynamics, envision resilient futures, include people and places); and Group 3 focussing on operaitonalising change on the ground (demonstrate pathway practically, enable agency of change, build transformative governance). For instance, in Group 3, a potential entry point for behaviour change intervention is to "support and create mechanisms for grassroots action" (Moallemi et al. 2025, p. 6). Targeting the mesosystem level, this intervention potentially allows grassroots actors to develop and strengthen their partnerships with their peers through relevant platforms (e.g. forums, conferences, technical advisory groups) at the microsystem level. Through participating in these activities, these individual actors can build their capability and motivation, and this may eventually result in them taking desirable behaviour change. In sum, behaviour change is the outcome of not just individual factors but also of chains of interdependent structures at different system levels.

6 Conclusion, limitations, and implications for further research

This paper has presented three case studies in which we applied tailored SeCOM-B models to help practitioners situate behaviour change problems including systemic drivers and barriers embedded in their socio-economic systems, and to design different interventions. In Case 1, we focussed on individual ethnic minority women and applied SeCOM-B to develop a behaviour change communication strategy to shift gender norms at the microsystem and mesosystem levels in Northern Vietnam. In Case 2, we focussed on individual farmers and used the model to identify the barriers and solutions at the microsystem (farmers' close networks) and mesosystem levels (farming community and economic sectors) associated with farmers' participation in environmental markets in Australia. In Case 3, we focussed on local councils as organisations (the microsystem) and council officers as individuals and used the model to co-design an intervention that helps local councils incorporate Dja Dja Wurrung people's knowledge into their climate adaptation planning in Victoria, Australia. The application of SeCOM-B across these three cases was flexible, depending on the unit of analysis (i.e. the case), target behaviours, and the broad contexts that constitute the systemic layers surrounding the behaviours.

Our study has some notable limitations. First, we applied SeCOM-B models in three case studies that centred on developing behaviour change interventions, yet, lacked the resources at the time of writing this paper to conduct a follow-up evaluation of the long-term effects and sustainability of these interventions. Such an evaluation outcome would strengthen the validity and reliability of our model. Second,

despite the insightful and rich information we collected in three different socio-economic and cultural contexts, the applicability of the SeCOM-B model needs further testing across cultures and drawing on different theoretical groundings. In this direction, we believe that combining this model with other critical theories, such as feminist geography, gender transformative approach, and intersectional analyses (Shields 2008; Hanson 2010), can avoid taken-for-granted biases and unchallenged ethical assumptions, generating more reliable findings on the feasibility and applicability of our model. Third, our research is case-based and qualitative in nature. Future research can utilise experimental approaches and randomised controlled trials to design and test the replicability and validity of the models in their settings. This could make their research findings more convincing to audiences less familiar with qualitative research.

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Kien Nguyen-Trung is an applied qualitative sociologist who integrates sociological imagination, systems thinking, behavioural science, and a range of qualitative research and co-creation methods to surface community insights and guide policy. His work advances climate change adaptation and disaster resilience by reducing inequities and social exclusion, while fostering sustainable transformation across Australia and the Greater Mekong region. He pioneers the use of generative AI and other novel research approaches to push the boundaries of qualitative inquiry.



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